

Exercise (1) $\Delta y/x = y = x + 30$ $\Delta x/y = x = \frac{1}{4}y + 60$.

1) correlation coefficient :

$$\left. \begin{aligned} a &= \frac{\text{cov}(x, y)}{\text{Var}(x)} = 1 \quad \text{--- (1)} \\ a' &= \frac{\text{cov}(x, y)}{\text{Var}(y)} = \frac{1}{4} \quad \text{--- (2)} \end{aligned} \right\} \Rightarrow \rho(x, y) = \frac{\text{cov}(x, y)}{\sqrt{\text{Var}(x) \times \text{Var}(y)}}$$

- you can calculate it directly.

$$\underline{r^2 = a \times a'}$$

* by multiplying (1) and (2) we get :

$$a \times a' = \frac{\text{cov}(x, y) \times \text{cov}(x, y)}{\text{Var}(x) \times \text{Var}(y)} = \frac{[\text{cov}(x, y)]^2}{s_x^2 \times s_y^2} \Rightarrow \left[\frac{\text{cov}(x, y)}{s_x \times s_y} \right]^2 = 1 \times \frac{1}{4}$$

$$\Rightarrow \frac{1}{4} = \left[\frac{\text{cov}(x, y)}{s(x) \times s(y)} \right]^2 \Rightarrow \frac{1}{2} = \frac{\text{cov}(x, y)}{s(x) \times s(y)} \Rightarrow \boxed{\rho(x, y) = \frac{1}{2}}$$

2) Arithmetic mean of the variable X and y :

$$\text{we have } y = x + 30 \Rightarrow \bar{y} = \bar{x} + 30 \quad \text{--- (1)}$$

$$x = \frac{1}{4}y + 60 \Rightarrow \bar{x} = \frac{1}{4}\bar{y} + 60 \quad \text{--- (2)}$$

* by compensation (1) in (2) :

$$\bar{x} = \frac{1}{4}(\bar{x} + 30) + 60 = \frac{\bar{x}}{4} + 7,5 + 60 \Rightarrow \bar{x} - \frac{\bar{x}}{4} = 67,5 \Rightarrow \boxed{\bar{x} = 90}$$

$$\bar{y} = \bar{x} + 30 = 90 + 30 \Rightarrow \boxed{\bar{y} = 120}$$

3) the covariance between X and y : $\text{var}(y) = [s(y)]^2$

$$a' = \frac{\text{cov}(x, y)}{s(y)^2} \Rightarrow \text{cov}(x, y) = a' \times s(y)^2 = \frac{1}{4} \times 40 \Rightarrow \boxed{\text{cov}(x, y) = 10}$$

Exercise (2) : speed (x) Distance (y).

1) the mean :

$$\bar{X} = \frac{\sum x_i}{n} = \frac{750}{10} \Rightarrow \boxed{\bar{x} = 75} \quad \left| \quad \bar{y} = \frac{\sum y_i}{n} = \frac{628}{10} \Rightarrow \boxed{\bar{y} = 62,8}$$

2) the variance :

$$\text{var}(x) = \frac{\sum x_i^2}{n} - (\bar{x})^2 = \frac{64500}{10} - (75)^2 \Rightarrow \boxed{\text{var}(x) = 825}$$

$$\text{var}(y) = \frac{\sum y_i^2}{n} - (\bar{y})^2 = \frac{57608}{10} - (62,8)^2 \Rightarrow \boxed{\text{var}(y) = 1816,96}$$

3) the standard deviation :

$$s(x) = \sqrt{\text{var}(x)} = \sqrt{825} = \boxed{28,72}$$

$$s(y) = \sqrt{\text{var}(y)} = \sqrt{1816,96} = \boxed{42,62}$$

4) the correlation coefficient:

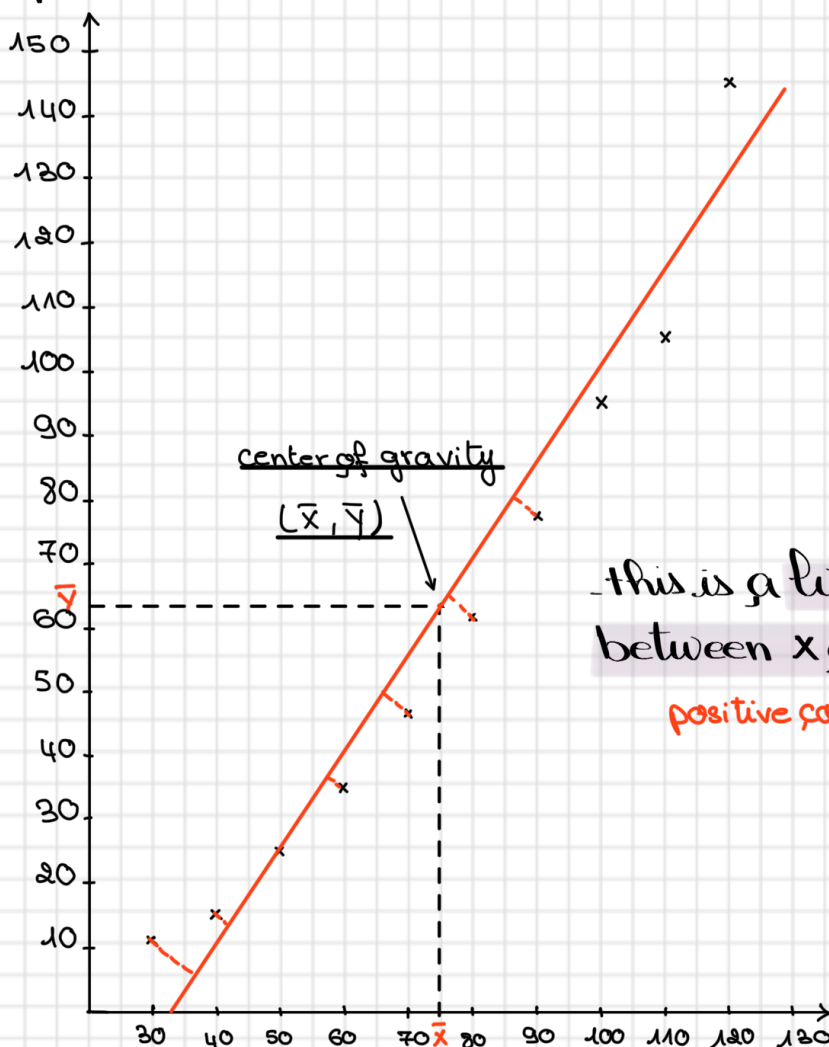
$$\rho(x, y) = \frac{\text{cov}(x, y)}{s(x) \times s(y)} \quad | \quad \text{cov}(x, y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{10} \times (-12000) = -1200$$

$$\hookrightarrow \rho(x, y) = \frac{-1200}{28,72 \times 42,62} \Rightarrow \rho(x, y) = -0,98$$

$$\text{or } \rho(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \times \sum (y_i - \bar{y})^2}} = \frac{-12000}{\sqrt{(2250) \times (12169,61)}} = -0,98$$

* it is a perfect positive linear correlation between x and y.

5) the scatter plot:



this is a linear regression between x and y (almost perfect positive correlation).

6) the regression line of y on x:

$$Dy/x: y = ax + b, \text{ where } a = \frac{\text{cov}(x, y)}{\text{Var}(x)} \text{ and } b = \bar{y} - a\bar{x}$$

$$\Rightarrow a = \frac{\text{cov}(x, y)}{\text{Var}(x)} = \frac{-1200}{825} = -1,45$$

$$\text{then } Dy/x: y = -1,45x - 45,95$$

$$\Rightarrow b = 62,8 - 1,45 \times 75 = -45,95$$

$$\text{at } 130 \text{ km/h } y = -1,45 \times 130 - 45,95 = -142,55 \text{ m}$$

7) the regression line fit the data closely.

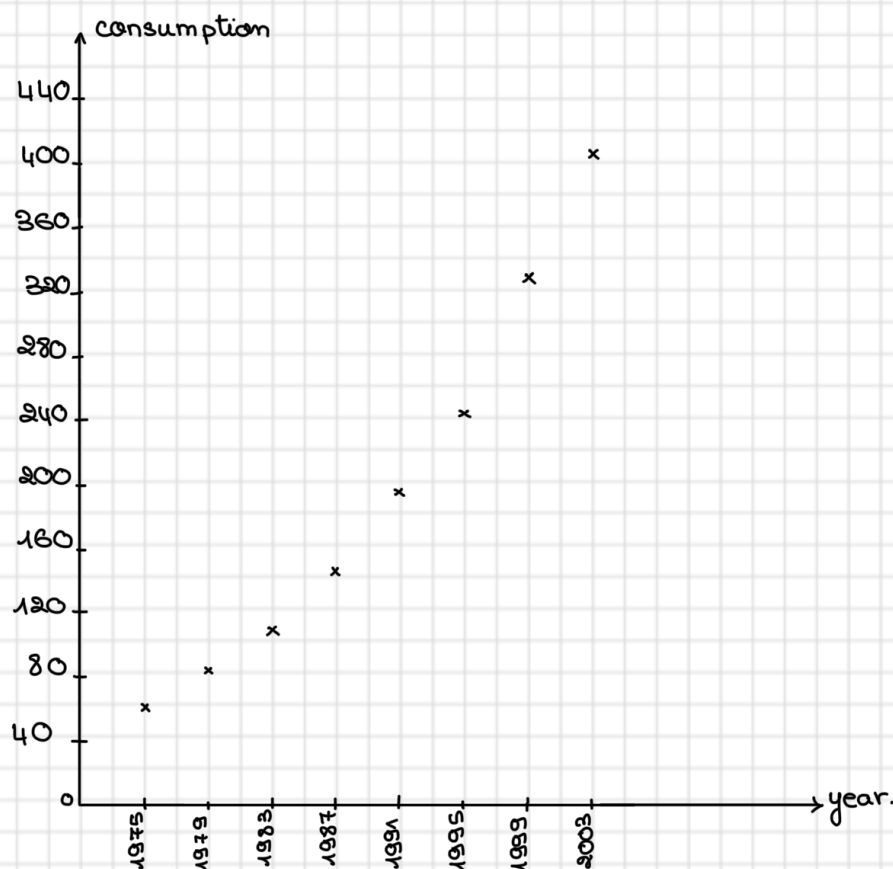
$$R^2 = \frac{VE(Y)}{Var(Y)} = \frac{VE(X)}{Var(X)} \Rightarrow \begin{cases} Var(Y) = VE(Y) + VR(Y) \\ VE(Y) = Var(Y) \cdot \rho^2(x, y) \\ VR(Y) = Var(Y) \cdot [1 - \rho^2(x, y)] \end{cases}$$

$$\Rightarrow R^2 = 1 - \frac{VR(Y)}{Var(Y)} = 1 - \frac{VR(X)}{Var(X)} \Rightarrow R^2 = \rho^2(x, y) \approx 0,9604 \approx 96,04\%$$

the regression line explain 96,05% of the variance of (y) and (x) which gives that the goodness of fit is very good.

Exercise (4):

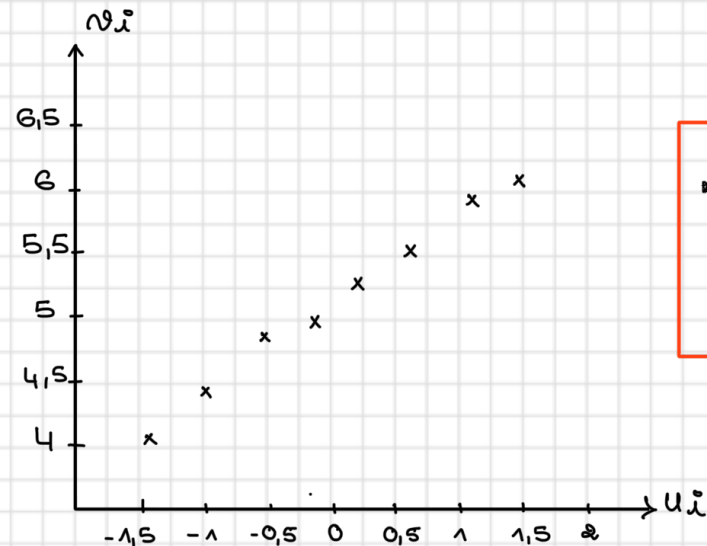
1) the scatter plot:



we have $u_i = \frac{x_i - 1989}{9,798}$ and $v_i = \log(y_i)$. So:

x_i	1975	1979	1983	1987	1991	1995	1999	2003
y_i	60	82	112	146	194	246	320	414
u_i	-1,43	-1,02	-0,61	-0,2	0,2	0,61	1,02	1,42
v_i	4,09	4,40	4,71	4,98	5,26	5,5	5,79	6,02

2) the scatter plot of transformed data:



* the purpose of the transformations seems to be linearizing the relationship between the variables

3) calculating $\bar{u}, \bar{v}, s^2 u, s^2 v, \text{cov}(u, v), \rho(u, v)$

$$\bar{u} = \frac{1}{n} \sum u_i = \frac{0}{8} = [0] \quad \left| \quad \bar{v} = \frac{1}{n} \sum v_i = [5,6938] \right.$$

$$s_u^2 = s^2 = \frac{1}{n} \sum_{i=1}^n (u_i - \bar{u})^2 = \frac{6,9378}{8} = [0,8672]$$

$$s_v^2 = s^2 = \frac{1}{n} \sum_{i=1}^n (v_i - \bar{v})^2 = \frac{3,1843}{8} = [0,3980]$$

$$\text{cov}(u, v) = \frac{\sum u_i v_i}{n} - \bar{u} \bar{v} = [0,58]$$

$$\rho(u, v) = \frac{\text{cov}(u, v)}{s_u \times s_v} = \frac{0,58}{\sqrt{0,8672 \times 0,3980}} = [0,9992]$$

4) the linear regression equation of v on u and measuring its goodness of fit.

$$D_{v/u}: v = a u + b \Rightarrow \begin{cases} a = \frac{\text{cov}(u, v)}{\text{Var}(u)} = [0,6769] \\ b = \bar{v} - a \bar{u} = [5,094] \end{cases}$$

then $D_{v/u}: v = 0,6769 u + 5,094$

* the coefficient of determination (R^2) is measure of the goodness of fit.

$$R^2 = \frac{VE(v)}{\text{Var}(v)} = \frac{0,3973}{0,3980} = [0,9984]$$

$$VE(v) = \text{Var}(v) \cdot \rho(u, v)^2 = 0,3980 \times 0,9992^2 = [0,3973]$$

- the goodness of fit is very good

5) the relation between u and v is perfect

6) the estimate of the consumption in the year 2007.

$$D_{v/u}: v = 0,6769 u + 5,094 \quad \left| \quad u_i = \frac{x_i - 1989}{9,798} \quad \right| \quad v_i = \log(y_i)$$

$$\log(y) = 0,6769 \left(\frac{x-1989}{9,798} \right) + 5,094 \Rightarrow y = e^{0,6769 \left(\frac{x-1989}{9,798} \right) + 5,094}$$

$$\Rightarrow y_{2007} = e^{0,6769 \left(\frac{2007-1989}{9,798} \right) + 5,094} \Rightarrow y_{2007} \approx \boxed{565,403}$$

Exercise (3) :